



SAINT IGNATIUS' COLLEGE

## Trial Higher School Certificate

2008

## MATHEMATICS EXTENSION 1

8:45am – 10:50am  
Friday 5<sup>th</sup> September

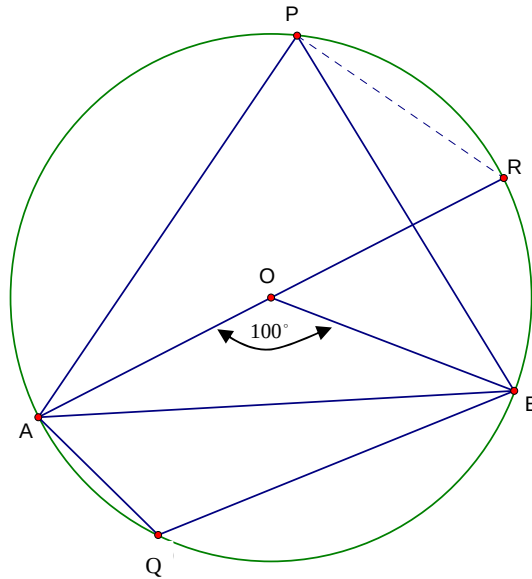
### Directions to Students

|                                                                                                        |                                    |
|--------------------------------------------------------------------------------------------------------|------------------------------------|
| • Reading Time: 5 minutes                                                                              | • Total Marks: 84                  |
| • Working Time: 2 hours                                                                                |                                    |
| • Write using blue or black pen (sketches in pencil).                                                  | • Attempt Questions 1 – 7          |
| • Board approved calculators may be used                                                               | • All questions are of equal value |
| • A table of standard integrals is provided at the back of this paper.                                 |                                    |
| • All necessary working should be shown in every question.                                             |                                    |
| • <b>Answer each question in the booklets provided and clearly label your name and teacher's name.</b> |                                    |

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QUESTION 1 (use a SEPARATE writing booklet)

- (a) If  $x = \frac{nx_1 + mx_2}{m+n}$  and  $x_1 = -2$  and  $x_2 = 3$ , find  $x$  when
- (i)  $m = 3, n = 2$  (1M)
- (ii)  $m = 3, n = -2$  (1M)
- (b) Noting that  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ , write down, without explanation, the value of:
- (i)  $\lim_{x \rightarrow 0} \frac{\sin 2008x}{x}$  (1M)
- (ii)  $\lim_{x \rightarrow 0} \frac{\sin(\pi - 2x)}{2008x}$ , assuming that  $\sin(\pi - 2x) = \sin 2x$ . (1M)
- (c) Write down the derivative, with respect to  $x$ , of
- (i)  $\cos^{-1} x$  (1M)
- (ii)  $e^{7x}$  (1M)
- (d) In the diagram, P, R, B, Q and A are points on the circumference of a circle centre O and AOR is a diameter.



- (i) Given  $\angle AOB = 100^\circ$ , briefly explain why  $\angle APB = 50^\circ$  (1M)
- (ii) Write down, without explanation, the size of
- (a)  $\angle AQB$  (1M)
- (b)  $\angle BPR$  (1M)

- (e) Noting that  $\frac{d}{dx}[e^x \sin x] = e^x \sin x + e^x \cos x$  evaluate

$$\int_0^{\frac{\pi}{2}} e^x (\sin x + \cos x) dx \quad (2M)$$

- (f) Given that if  $y = 7^x$ , then  $\frac{dy}{dx} = A(7^x)$ , where  $A$  is a constant,

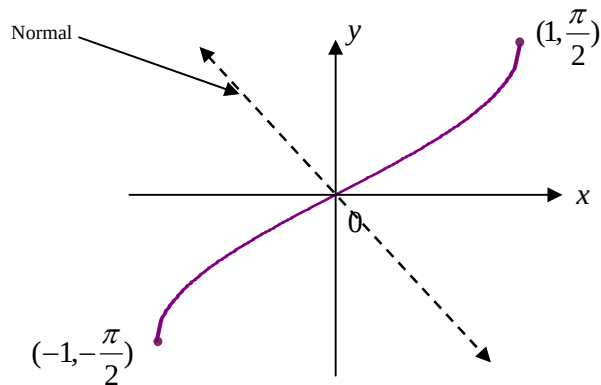
write down the exact value of  $A$ . (1M)

**QUESTION 2** (use a SEPARATE writing booklet)

- (a) Write down the fifth term in the expansion by the binomial theorem of
- (i)  $(a + b)^8$  (1M)

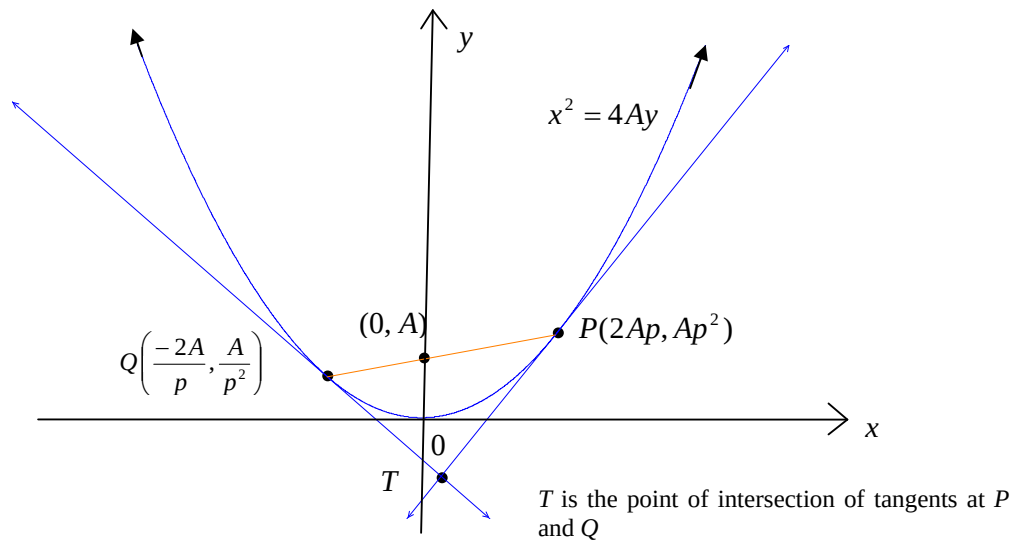
(ii)  $(3x + \frac{1}{x})^8$  (2M)

- (b) Focus on the graph of  $y = \sin^{-1} x$  below. State (answers only are required)



- (i) The domain of  $x$ . (1M)
- (ii) The gradient of the normal at O the origin. (1M)
- (iii) The equation of the horizontal line in which the curve  $y = \sin^{-1} x$  may be reflected, to obtain a sketch of  $y = \cos^{-1} x$ . (1M)
- (c) Note that if  $-\pi \leq \theta \leq \pi$ , the solution to  $\cos \theta = \frac{1}{2}$  is  $\theta = \pm \frac{\pi}{3}$ .  
Assuming the above, write down in radians, the general solution of the equation  $\cos \theta = \frac{1}{2}$ . (2M)

(d)



PQ is a focal chord of the parabola

$$x^2 = 4Ay$$

The tangent at P has equation  $y - px + Ap^2 = 0 \dots\dots\dots(\alpha)$ .

The tangent at Q has equation  $y + \frac{x}{p} + \frac{A}{p^2} = 0 \dots\dots\dots(\beta)$ .

- (i) Do not prove the above but solve the equations  $(\alpha)$  and  $(\beta)$  to show that the x- coordinate of T is  $A(p - \frac{1}{p})$ . (2M)
- (ii) Determine the y- coordinate of T. (2M)

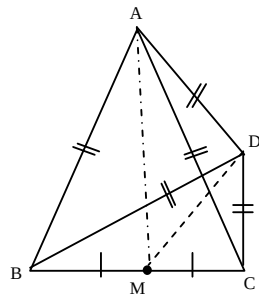
**QUESTION 3** (use a SEPARATE writing booklet)

- (a) Consider the polynomial  $P(x) = x^3 + Ax^2 - 2008$ .

If  $(x-2)$  is a factor of  $P(x)$  find  $A$ . (2M)

- (b) Use the principle of mathematical induction to show that  $(2008^n - 1)$  is divisible by 9 for all positive integers  $n$ . (4M)

- (c) The figure below ABCD is a regular tetrahedron, with

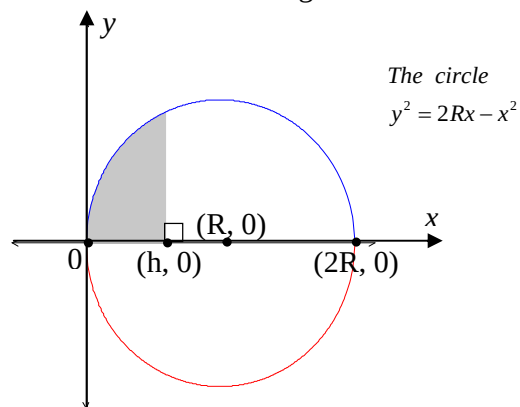


$AB = AC = BC = BD = DC = AD = 20\text{cm}$ .

- (i) Draw the triangle BCD in your writing booklet and mark 'M' the midpoint of BC. Prove that  $DM = 10\sqrt{3}\text{cm}$ . (2M)

- (ii) Determine the size of  $\hat{AMD}$ . (2M)

- (d) A circle centre  $(R, 0)$  and radius  $R$  has equation  $y^2 = 2Rx - x^2$  and is drawn below. The shaded portion of the circle is rotated through  $2\pi$  radians about the  $x$ -axis.



Prove that the volume  $V$  of the spherical cap generated is given by

$$V = \pi h^2 \left(R - \frac{h}{3}\right). \quad (2M)$$

**QUESTION 4** (use a SEPARATE writing booklet)

- (a) A point P moves along the parabola whose equation is  $y = \frac{x^2}{8}$ .  
The  $x$ -coordinate of P increases at the constant rate of 8cm/sec.  
At what rate is the  $y$ -coordinate increasing when  $x = 4$ ? (2M)

- (b) In solving a problem on Newton's Law of Cooling, Ignatius correctly arrives at the equation.

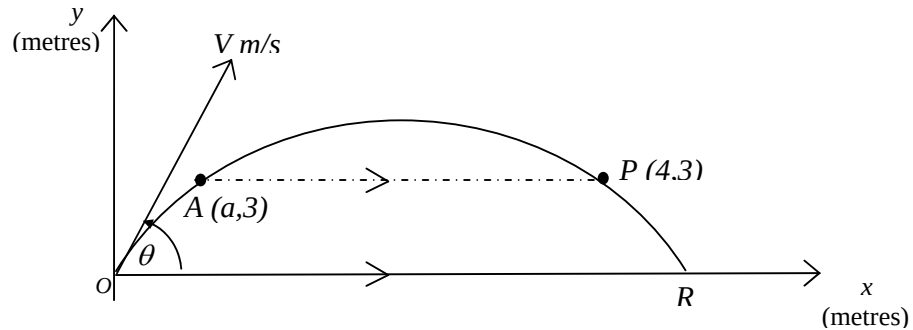
$$T = 20 + 980e^{-kt}$$

Where  $T$  is the temperature of a body at time  $t$ .

- (i) Given that at  $t = 100$ ,  $T = 510$ , show  $k = \frac{\ln 2}{100}$ . (1M)
- (ii) Find  $T$  when  $t = 200$ . (1M)
- (c) If the equation  $f(x) = 0$  has a root near  $x = a$ , it can be shown that, in general, a closer approximation to the root is  $a_1$ , where  $a_1 = a - \frac{f(a)}{f'(a)}$ . Do not prove this result, known as Newton's method of approximating roots, but draw and label:
- (i) One diagram showing the regular situation in which  $a_1$  is a better approximation than  $a$ . (2M)
- (ii) A second diagram, showing a case in which Newton's method does not apply, even though  $f(x)$  has a root near  $x = a$ . (1M)

Question 4 continues on the next page.

- (d) A particle is projected from a point  $O$  with a speed of  $V$  m/s at an angle of  $\theta$  to the horizontal. Air resistance is to be neglected and  $g$  ms<sup>-2</sup> is the acceleration due to gravity, vertically downwards.



In solving the following problems, you may assume that the Cartesian equation of the path of the projectile is given by:

$$y = \frac{-gx^2}{2V^2}(1 + \tan^2 \theta) + x \tan \theta. \text{ (Do NOT prove this result)}$$

You are given  $V^2 = 8g$  and that the particle passes through a point  $P(4,3)$ .

- (i) Prove that  $\tan \theta = 2$ , without using the result of (ii) below. (2M)
- (ii) Assume that  $\tan \theta = 2$ , to show that the range  $OR = 6.4$  m. (2M)
- (iii) The point  $A = (a, 3)$  lies on the trajectory, where  $AP$  is parallel to the x-axis. What is the value of  $a$ ? (1M)

QUESTION 5 (use a SEPARATE writing booklet)

- (a) Noting that  $\log_e e^N = N$  show by setting  $u = \log_e x$ , that (3M)

$$2 \int_{e^2}^{e^5} \frac{\log_e x}{x} dx = 21$$

- (b) Find the values of  $x$  and  $y$  which satisfy the simultaneous equations

$$\cos^{-1} x + 3 \sin^{-1} y = \pi \dots\dots(\alpha) \quad (3M)$$

$$\cos^{-1} x - 3 \sin^{-1} y = 0 \dots\dots(\beta)$$

- (c) Here you may use, without proof, the identity

$$\sin 3A = 3 \sin A - 4 \sin^3 A \dots\dots(\alpha)$$

- (i) By direct substitution show  $\alpha$  is true if  $A = \frac{\pi}{6}$ . (1M)

- (ii) Use  $\alpha$  to find, in exact form, the smallest positive value of  $x$

$$\text{which satisfies the equation } 6x - 8x^3 = 1. \quad (3M)$$

(suggestion: let  $x = \sin A$ )

- (d) Assuming that

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \dots\dots\delta$$

factorise

$$(\alpha - \beta)^3 + (\beta - \gamma)^3 + (\gamma - \alpha)^3. \quad (2M)$$

**QUESTION 6** (use a SEPARATE writing booklet)

- (a) A particle, executing Simple Harmonic Motion, is moving along the  $x$ -axis with velocity  $v$ , and acceleration  $\ddot{x}$ .

(i) Show that  $\ddot{x} = \frac{d}{dx} \left( \frac{v^2}{2} \right)$ . (1M)

- (ii) Assuming that in the usual notation  $\ddot{x} = -n^2 x$  and that the particle is released from rest from the point where  $x = a$ , and at zero time, show that:

( $\alpha$ )  $v = -n\sqrt{a^2 - x^2}$ . (2M)

( $\beta$ )  $x = a \cos nt$ . (2M)

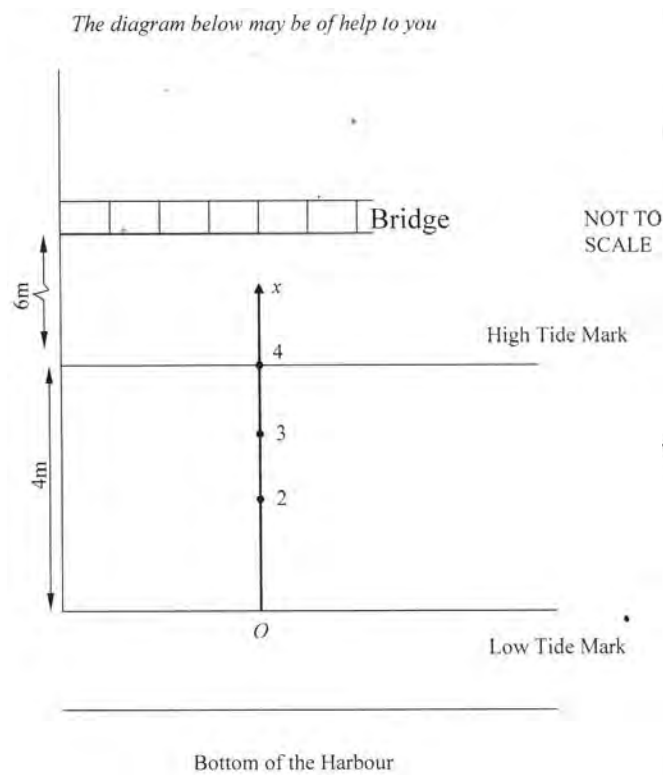
- (b) In Aloha Harbour the height of the water above the low tide mark is given by the rule  $x = 2(1 + \cos \theta)$  metres, where  $\theta = \frac{\pi t}{390}$  radians and  $t$  is the time in minutes after high tide. (see the diagram on the next page)

Explain why the time interval between high and low tide is 390 minutes. (2M)

- (c) Given  $t = 65$ , show that the rate at which the water level is falling is approximately 8mm per minute. (2M)

Question 6 continues on the next page.

(d)



The Aloha Harbour Bridge is 10 metres above the low water mark.

A pleasure yacht can just sail under the bridge when the distance between the bridge and the water is 7 metres. How long after high tide will it be before the yacht can first sail under the bridge?

(3M)

QUESTION 7 (use a SEPARATE writing booklet)

- (a) You may assume that  $x^4(1-x)^4 = x^8 - 4x^7 + 6x^6 - 4x^5 + x^4$

Let  $J = \int_0^1 x^4(1-x)^4 dx$ .

Show that  $J = \frac{1}{630}$  (3M)

- (b) Let  $E = \frac{x^4(1-x)^4}{x^2+1}$ .

Noting the assumption in part (a) and using the process of long division for polynomials, show that

$$E = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2} \quad (3M)$$

- (c) Let  $I = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ . Use the result in part (b) to prove that (3M)

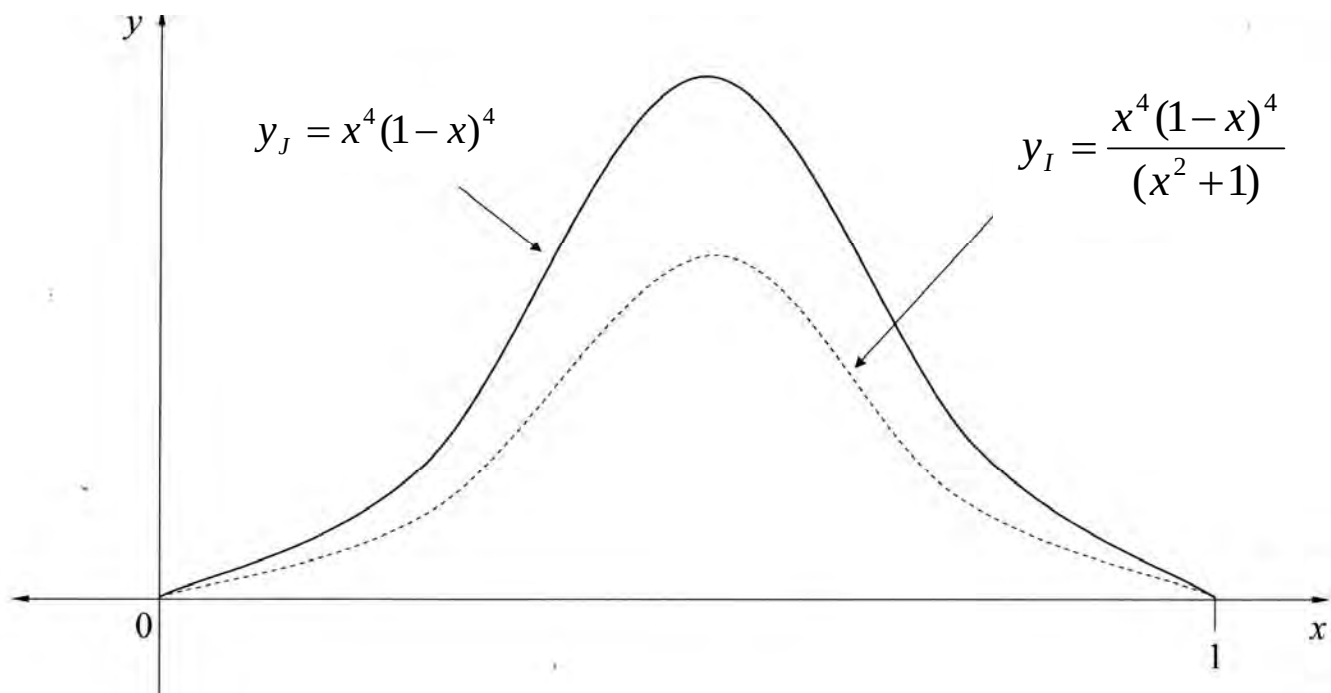
$$I = \frac{22}{7} - \pi.$$

- (d) Assuming, without any explanation, that  $J > I > \frac{J}{2}$ , deduce from (a) and (c) that

$$\frac{22}{7} - \frac{1}{630} < \pi < \frac{22}{7} - \frac{1}{1260}. \quad (2M)$$

Question 7 continues on the next page.

(e)



Use the diagram above, or otherwise, to explain why  $J > I$ .

(1M)



SAINT IGNATIUS' COLLEGE

Suggested  
Solutions

## Trial Higher School Certificate

2008

# MATHEMATICS EXTENSION 1

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Friday 5<sup>th</sup> September

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Solutions, Marking Scheme & Comments

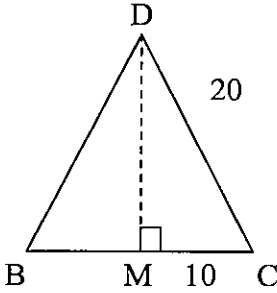
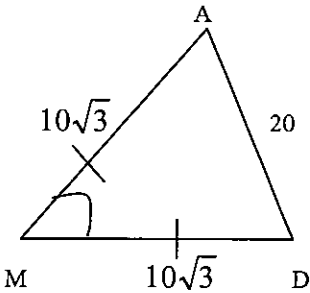
| QUESTION 1                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |           |                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|-------------------|
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                        | Max Mark                   | Your Mark | Marker's Comments |
| <p>(a)</p> <p>(i) <math>x_{(i)} = \frac{2 \times (-2) + (3 \times 3)}{(3 + 2)} = \frac{5}{5} = 1</math></p> <p>(ii) <math>x_{(ii)} = \frac{(-2 \times -2) + (3 \times 3)}{(3 - 2)} = \frac{13}{1} = 13</math></p> <p><i>(Award the mark if the candidate makes just one error)</i></p>                                                                                                                                                     | <p>1</p> <p>1</p>          |           |                   |
| <p>(b)</p> <p>(i) <math>E_{(i)} = (2008) \lim_{x \rightarrow 0} \left( \frac{\sin 2008x}{2008x} \right) = 2008 \times 1 = 2008</math></p> <p>(ii) <math>E_{(ii)} = \lim_{x \rightarrow 0} \left( \frac{\sin 2x}{2x} \right) \times \frac{1}{1004} = \frac{1}{1004}</math></p>                                                                                                                                                              | <p>1</p> <p>1</p>          |           |                   |
| <p>(c)</p> <p>(i) <math>\frac{-1}{\sqrt{1-x^2}},  x  \leq 1</math></p> <p>(ii) <math>7e^{7x}</math></p>                                                                                                                                                                                                                                                                                                                                    | <p>1</p> <p>1</p>          |           |                   |
| <p>(d)</p> <p>(i) In the given circle, <math>\hat{AOB}</math> is the angle at the centre, subtended by the arc <math>AQB</math>. <math>\hat{APB}</math> is the angle at the circumference of the same circle, subtended by the arc <math>AQB</math>.</p> <p><math>\therefore \hat{APB} = \frac{1}{2} \hat{AOB} = \frac{100^\circ}{2} = 50^\circ</math>.</p> <p>(ii)</p> <p>(a) <math>130^\circ</math></p> <p>(b) <math>40^\circ</math></p> | <p>1</p> <p>1</p> <p>1</p> |           |                   |
| <p>(e)</p> <p><math>I = \left[ e^x \sin x + e^x \cos x \right]_0^{\frac{\pi}{2}}</math></p> <p><math>= \left[ e^{\frac{\pi}{2}} + 0 \right] - [0 + 1]</math></p> <p><math>= e^{\frac{\pi}{2}} - 1</math></p> <p><i>(No penalty for one error)</i></p>                                                                                                                                                                                      | <p>2</p>                   |           |                   |
| <p>(f) <math>A = \ln 7</math></p>                                                                                                                                                                                                                                                                                                                                                                                                          | <p>1</p>                   |           |                   |

| QUESTION 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |           |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|-------------------|
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Max Mark                   | Your Mark | Marker's Comments |
| <p>(a)</p> <p>(i) <math>T_5 = \binom{8}{4} a^4 b^4</math></p> <p>(ii) <math>T_5 = \binom{8}{4} (3x)^4 \left(\frac{1}{x}\right)^4 = \frac{8.7.6.5}{4.3.2.1} 81.x^0 = 5670</math><br/>(in part (ii) no penalty for one error)</p>                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>1</p> <p>2</p>          |           |                   |
| <p>(b)</p> <p>(i) <math>-1 \leq x \leq 1</math></p> <p>(ii) <math>-1</math></p> <p>(iii) <math>y = \frac{\pi}{4}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>1</p> <p>1</p> <p>1</p> |           |                   |
| <p>(c)</p> <p>General Solution is <math>\theta = 2\pi n \pm \frac{\pi}{3}</math> where n is an integer, positive, negative or zero.<br/>Award 1 mark for giving some new correct value of <math>\theta</math> in radians.</p>                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                          |           |                   |
| <p>(d)</p> <p>(i) Eliminate <math>y</math> to get <math>x\left(p + \frac{1}{p}\right) - A\left(p^2 - \frac{1}{p^2}\right) = 0</math><br/>So <math>x\left(p + \frac{1}{p}\right) = A\left(p^2 - \frac{1}{p^2}\right)</math><br/><math display="block">= A\left(p - \frac{1}{p}\right)\left(p + \frac{1}{p}\right)</math><math display="block">\therefore x = A\left(p - \frac{1}{p}\right)</math></p> <p>(ii) Substitute <math>x = A\left(p - \frac{1}{p}\right)</math> into <math>y = px - Ap^2</math> to get<br/><math display="block">y = p\left[Ap - \frac{A}{p}\right] - Ap^2 = Ap^2 - A - Ap^2 = -A</math><br/>The y coordinate of T is -A</p> | <p>2</p> <p>2</p>          |           |                   |

QUESTION 3

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Max Mark | Your Mark | Marker's Comments |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(a)</p> <p>Let <math>f(x) = x^3 + Ax^2 - 2008</math> then <math>f(2) = 0</math><br/> <math>\therefore 0 = 8 + 4A - 2008 \therefore A = 500</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2        |           |                   |
| <p>(b)</p> <p><b>Step 1:</b></p> <p>When <math>n = 1</math>, <math>2008^1 - 1 = 2007 = 223 \times 9</math>,<br/>         Which is divisible by 9, and<br/>         Thus the statement is true for <math>n = 1</math>.</p> <p><b>Step 2:</b></p> <p>Assume the statement true for <math>n = k</math><br/>         i.e. assume that <math>2008^k - 1</math> is divisible by 9,<br/>         i.e. that <math>\frac{2008^k - 1}{9}</math> is some integer <math>M</math>.<br/>         i.e. <math>\frac{2008^k - 1}{9} = M</math>,<br/>         i.e. <math>2008^k - 1 = 9M \Rightarrow 2008^k = 9M + 1 \dots \dots **</math></p> <p>We now prove that the statement is true for <math>n = k + 1</math><br/>         i.e. <math>2008^{k+1} - 1</math> is divisible by 9</p> <p>Now <math>2008^{k+1} - 1 = 2008^1 \cdot 2008^k - 1</math><br/> <math>= 2008(9M + 1) - 1 \dots \dots **(\text{by our assumption})</math><br/> <math>= 9 \times 2008M + 2008 - 1</math><br/> <math>= 9 \times 2008M + 2007</math><br/> <math>= 9(2008M + 223)</math><br/> <math>= 9I</math> (where <math>I</math> is a positive integer)<br/>         which is divisible by 9</p> <p>Thus, if <math>2008^k - 1</math> is divisible by 9 when <math>n = k</math>,<br/>         then it is also divisible by 9 when <math>n = k + 1</math></p> <p><b>Step 3:</b></p> <p>Since the statement is true for <math>n = 1</math>, it is true for <math>n = 1 + 1 = 2</math>, and<br/> <math>n = 2 + 1 = 3</math>, and so on for all positive integral values of <math>n</math>.</p> | 4        |           |                   |

QUESTION 3

| Suggested Solutions                                                                                                                                                                                                                                                                                                                    | Max Mark | Your Mark | Marker's Comments |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(c) (i)</p>  <p>From symmetry, <math>\hat{DMC} = 90^\circ</math><br/>In right angle triangle <math>DMC</math><br/><math>DM^2 = 20^2 - 10^2 = 300</math><br/><math>= 100 \times 3</math><br/><math>\therefore DM = 10\sqrt{3} \text{ cm}</math></p> | 2        |           |                   |
| <p>(c) (ii)</p>  <p>Apply cosine rule to <math>\triangle AMD</math>.<br/><math>\cos \hat{AMD} = \frac{300 + 300 - 400}{600} = \frac{1}{3}</math><br/><math>\therefore \hat{AMD} = \cos^{-1}\left(\frac{1}{3}\right) \approx 70^\circ 32'</math></p>  | 2        |           |                   |
| <p>(d)</p> $V = \pi \int_0^h (2Rx - x^2) dx$ $= \pi \left[ Rx^2 - \frac{x^3}{3} \right]_0^h$ $= \pi \left[ Rh^2 - \frac{h^3}{3} \right] - [0]$ $= \pi h^2 \left[ R - \frac{h}{3} \right]$                                                                                                                                              | 2        |           |                   |

(a)

When  $x = 4$ ,  $\frac{dy}{dt} = \left( \frac{4}{4} \times 8 \right) = 8 \text{ cm/sec}$

2

### Marker's Comments

(i) Substitute values given to get  $510 = 20 + 980e^{-100K}$

$$\therefore e^{-100K} = \frac{490}{980} = \frac{1}{2} \longrightarrow \therefore K = \frac{\ln 2}{100}$$

1

(ii) Here  $T = 20 + 980e^{-2 \ln 2}$

$$= 20 + 245$$

$$= 265$$

1

(i) See Page 14.

2

(ii) See Pages 15 and 16.

1

(i) Into the given equation, substitute  $V^2 = 8g$ ,  $x = 4$ ,  $y = 3$  (data) to get

$$3 = \frac{-16g}{2 \times 8g} (1 + \tan^2 \theta) + 4 \tan \theta$$

$$3 = -1 - \tan^2 \theta + 4 \tan \theta$$

So  $\tan^2 \theta - 4 \tan \theta + 4 = 0$

$$\therefore (\tan \theta - 2)^2 = 0 \text{ i.e. } \tan \theta = 2 \text{ (twice)}$$

**Note.** That no other value of  $\tan \theta$  satisfies.

2

(ii) Set  $y = 0$ , to get

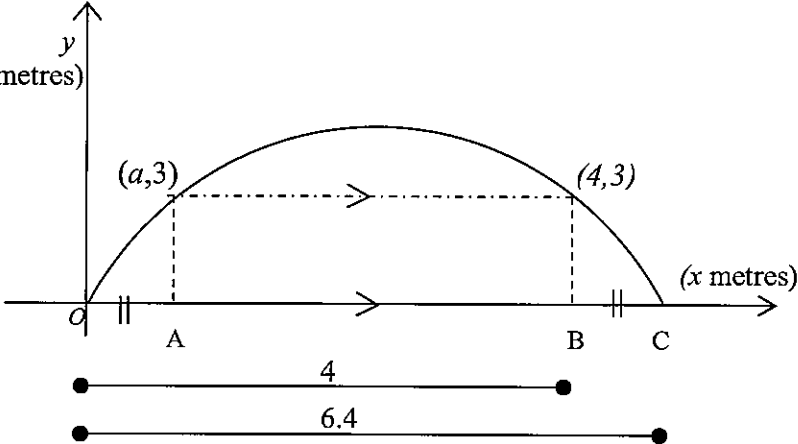
$$0 = \frac{-x^2}{16}(1+4) + 2x$$

$$\therefore \frac{x}{16}(5x-32)=0$$

$$\therefore x = 0 \text{ or } x=6.4$$

2

QUESTION 4

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                      | Max Mark | Your Mark | Marker's Comments |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(d)</p> <p>(iii)</p>  <p>From symmetry, <math>OA = a = BC</math><br/> Now <math>OC = 6.4</math><br/> <math>BC = OC - OB</math><br/> <math>= 6.4 - 4</math><br/> <math>= 2.4</math><br/> <math>\therefore OA = a = BC</math><br/> <math>\therefore a = 2.4</math></p> | 1        |           |                   |

[illegible]

QUESTION 5

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Max Mark | Your Mark | Marker's Comments |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(d) Given,</p> $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \dots (\delta)$ <p>Let <math>a = (\alpha - \beta), b = (\beta - \gamma), c = (\gamma - \alpha)</math> in the given identity <math>\delta</math>.</p> <p>Then <math>a + b + c = (\alpha - \beta) + (\beta - \gamma) + (\gamma - \alpha) = 0</math>, so R.H.S of <math>\delta = 0</math>.</p> $\therefore a^3 + b^3 + c^3 - 3abc = 0$ $\therefore a^3 + b^3 + c^3 = 3abc$ <p>ie <math>(\alpha - \beta)^3 + (\beta - \gamma)^3 + (\gamma - \alpha)^3 = 3(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)</math></p> <p><math>\therefore</math> The required factors are <math>3(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)</math></p> | 2        |           |                   |

QUESTION 6

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Max Mark | Your Mark | Marker's Comments |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(a)</p> <p>(i) Many approaches are possible for example</p> $RHS = \frac{1}{2} \frac{d}{dx}(v^2) = \frac{1}{2} 2v \frac{dv}{dx}$ $= \frac{dx}{dt} \cdot \frac{dv}{dx} = \frac{dv}{dt} = \text{acceleration}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1        |           |                   |
| <p>(ii) (<math>\alpha</math>)</p> <p>From part (i) <math>\frac{d}{dx}\left(\frac{v^2}{2}\right) = -n^2 x</math>, (where <math>n = \frac{2\pi}{T}</math>, T being the period). Integrate both sides with respect to x to get,</p> $\frac{v^2}{2} = -\frac{n^2 x^2}{2} + B. \text{ (B the constant of integration)}$ <p>From data, when <math>t = 0</math>, <math>v = 0</math> and <math>x = a</math></p> $\therefore 0 = -\frac{n^2 a^2}{2} + B \Rightarrow B = \frac{n^2 a^2}{2} \Rightarrow \frac{v^2}{2} = -\frac{n^2 x^2}{2} + \frac{n^2 a^2}{2}.$ <p>Multiply by 2 to arrive at ,</p> $v^2 = n^2 (a^2 - x^2), \quad \text{so } v = \pm n\sqrt{a^2 - x^2}.$ <p>But when <math>t = 0</math>, <math>x = a</math> (data), so particle must begin by moving to the left. (Note <math>x = a</math> is the maximum positive position of P).</p> $\therefore v = -n\sqrt{a^2 - x^2}, \quad  x  \leq a$ | 2        |           |                   |
| <p>(ii) (<math>\beta</math>)</p> <p>We can re-arrange the result of part (ii) (<math>\alpha</math>)</p> <p>viz, <math>\frac{dx}{dt} = -n\sqrt{a^2 - x^2}</math>, <math> x  \leq a</math>, to get <math>-\int \frac{dx}{\sqrt{a^2 - x^2}} = \int n dt</math></p> $\therefore \cos^{-1}\left(\frac{x}{a}\right) = nt + C. \text{ (C is the constant of integration).}$ <p>But from data, when <math>t = 0</math>, <math>x = a</math>. <math>\therefore C = 0</math>, as <math>\cos^{-1}(1) = 0</math></p> $\therefore \cos^{-1}\left(\frac{x}{a}\right) = nt \Rightarrow \text{So } x = a \cos nt$                                                                                                                                                                                                                                                                                                   | 2        |           |                   |

QUESTION 6

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                             | Max Mark | Your Mark | Marker's Comments |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(b) Here <math>x = 2 + 2 \cos\left(\frac{\pi}{390}\right)t</math>, the standard S.H.M. form</p> $x = a \cos nt. \text{ So } n = \left(\frac{\pi}{390}\right) \text{ and period} = \frac{2\pi}{n} = 780 \text{ mins}$ <p>So from high tide to next high tide = 780 mins.</p> <p>Hence from high tide to next low tide = 390 mins.</p>                         | 2        |           |                   |
| <p>(c) When <math>t = 65</math>,</p> $\frac{dx}{dt} = -\frac{\pi}{195} \sin \frac{\pi t}{390} = -\frac{\pi}{195} \sin \frac{\pi 65}{390} = \left(-\frac{\pi}{390} 1000\right) \text{ mm/min}$ $\approx -8.055...(\text{calc}) \approx -8 \text{ mm/min.}$ <p>The minus sign indicates that the water level is falling.</p>                                      | 2        |           |                   |
| <p>(d) Since the distance between the bridge and the water has to be 7m, clearly <math>x=3</math></p> $\therefore 2(1 + \cos \frac{\pi t}{390}) = 3, \text{ giving } \cos \frac{\pi t}{390} = \frac{1}{2}$ $\therefore \frac{\pi t}{390} = \frac{\pi}{3}. \text{ Hence } t = 130$ <p>The yacht can travel under the bridge 2 hours 10 mins after high tide.</p> | 3        |           |                   |

QUESTION 7

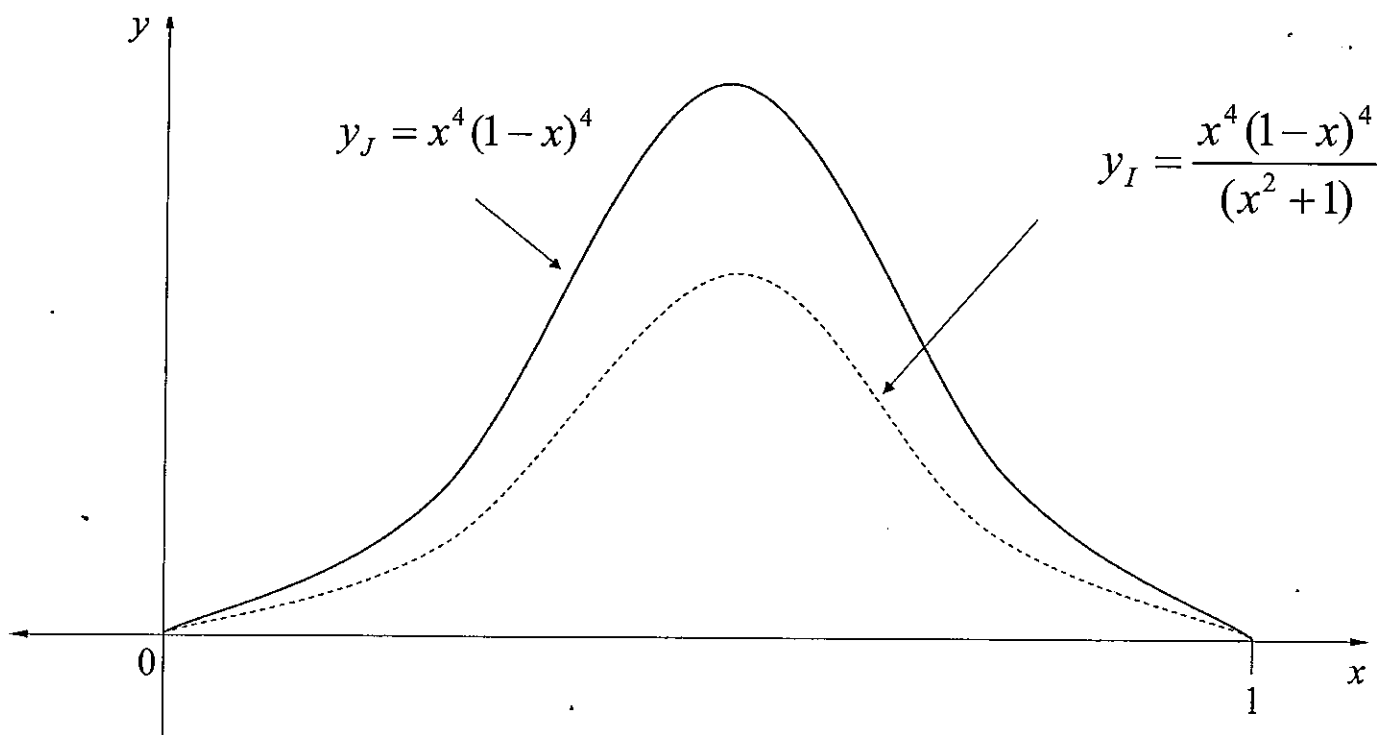
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Max Mark | Your Mark | Marker's Comments |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(a)</p> $J = \int_0^1 x^4(1-x)^4 dx$ $= \int_0^1 (x^8 - 4x^7 + 6x^6 - 4x^5 + x^4) dx \dots\dots\dots (given)$ $= \left[ \frac{x^9}{9} - \frac{x^8}{2} + \frac{6x^7}{7} - \frac{4x^6}{6} + \frac{x^5}{5} \right]_0^1$ $= \frac{1}{9} - \frac{1}{2} + \frac{6}{7} - \frac{2}{3} + \frac{1}{5}$ $= \frac{70 - 315 + 540 - 420 + 126}{630}$ $= \frac{736 - 735}{630} = \frac{1}{630}$                                                                                                                                                                                                                                                             | 3        |           |                   |
| <p>(b) First do the long division</p> $\begin{array}{r} x^6 - 4x^5 + 5x^4 - 4x^2 + 4 \\ x^2 + 1 \overline{) x^8 - 4x^7 + 6x^6 - 4x^5 + x^4} \\ \underline{x^8 \phantom{- 4x^7} + x^6} \phantom{- 4x^5 + x^4} \\ -4x^7 + 5x^6 - 4x^5 + x^4 \\ \underline{-4x^7 \phantom{+ 5x^6} - 4x^5} \phantom{+ x^4} \\ +5x^6 \phantom{- 4x^5} + x^4 \\ \underline{+5x^6 \phantom{- 4x^5} + 5x^4} \phantom{+ x^4} \\ -4x^4 \phantom{+ x^4} \\ \underline{-4x^4 - 4x^2} \phantom{+ x^4} \\ 4x^2 \phantom{+ x^4} \\ \underline{4x^2 + 4} \phantom{+ x^4} \\ -4 \phantom{+ x^4} \end{array}$<br>$\therefore E = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{x^2 + 1}$ | 3        |           |                   |

QUESTION 7

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Max Mark | Your Mark | Marker's Comments |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(c)</p> $I = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ $= \int_0^1 (x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}) dx \dots (\text{from part b})$ $= \left[ \frac{x^7}{7} - \frac{4x^6}{6} + x^5 - \frac{4x^3}{3} + 4x - 4 \tan^{-1} x \right]_0^1$ $= \left[ \frac{1}{7} - \frac{2}{3} + \frac{5}{5} - \frac{4}{3} + 4 \right] - 4[\tan^{-1}(1)]$ $= \frac{30 - 140 + 210 - 280 + 840}{210} - 4\left(\frac{\pi}{4}\right)$ $= \frac{1080 - 420}{210} - \pi$ $= \frac{660}{210} - \pi$ $= \frac{22}{7} - \pi$ | 3        |           |                   |
| <p>(d)</p> <p>Given that</p> $J > I > \frac{J}{2}$ $\frac{1}{630} > \frac{22}{7} - \pi > \frac{1}{1260} \text{ from parts (a) and (c)}$ <p>(multiply through by -1)</p> $-\frac{1}{630} < \pi - \frac{22}{7} < -\frac{1}{1260}$ <p>(add <math>\frac{22}{7}</math> to each expression)</p> $\frac{22}{7} - \frac{1}{630} < \pi < \frac{22}{7} - \frac{1}{1260}$                                                                                                                                                | 2        |           |                   |

QUESTION 7

| Suggested Solutions                                                                                                                                                                          | Max Mark | Your Mark | Marker's Comments |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------------------|
| <p>(e) From observation <math>J</math> represents the area under the full curve.<br/><math>I</math> represents the area under the broken curve.</p> <p>So clearly <math>J &gt; I</math>.</p> | 1        |           |                   |



An Algebraic Proof:

For  $0 < x < 1$  we have

$$1 < 1 + x^2 < 2$$

$$\text{So } x^4(1-x)^4 > \frac{x^4(1-x)^4}{1+x^2} > \frac{x^4(1-x)^4}{2}$$

$$J > I > \frac{J}{2}$$

Source of Question 7:

*N.S.W. Leaving Certificate Maths 1 Honours Paper 1964, Q9.*

*Also adapted from a Cambridge Matriculation examination paper, late 1930's.*

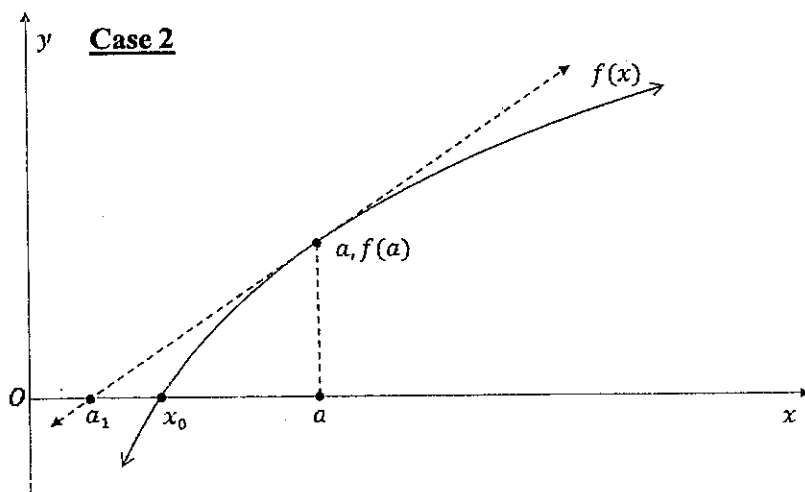
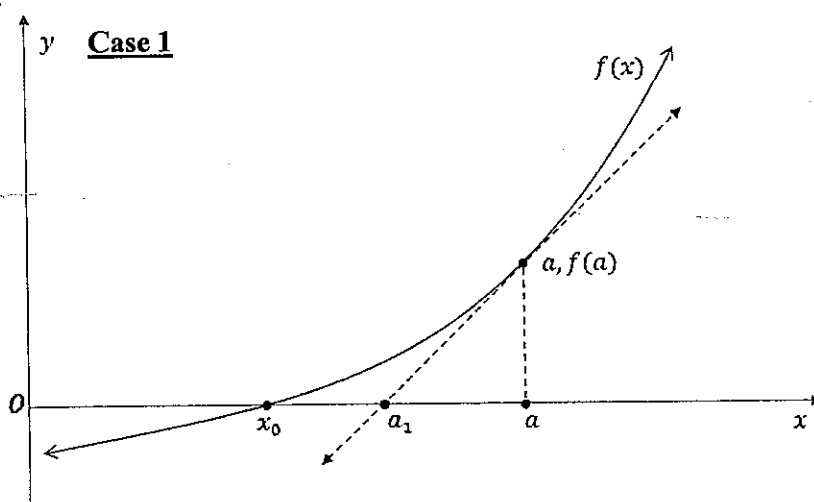
**Question (4C)**

In the following examples the required root of  $f(x)=0$  is given by  $x_0$ .

In cases 1 and 2 below, Newton's method works and  $a_1$  is a better approximation than  $a$ .

(i)

(2 marks)

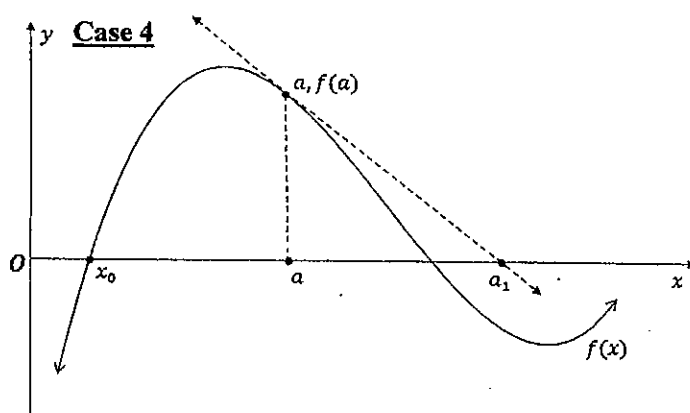
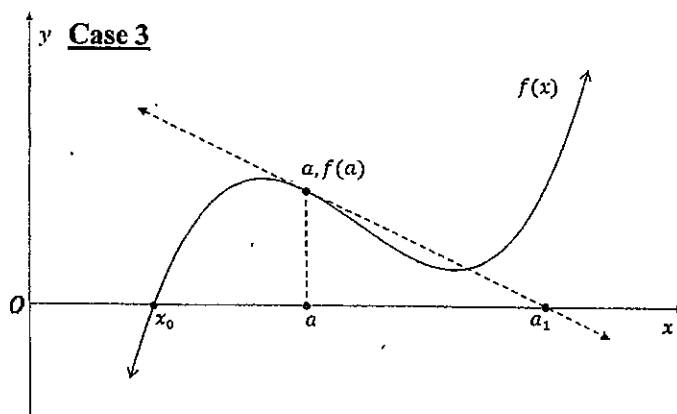


Question (4C)

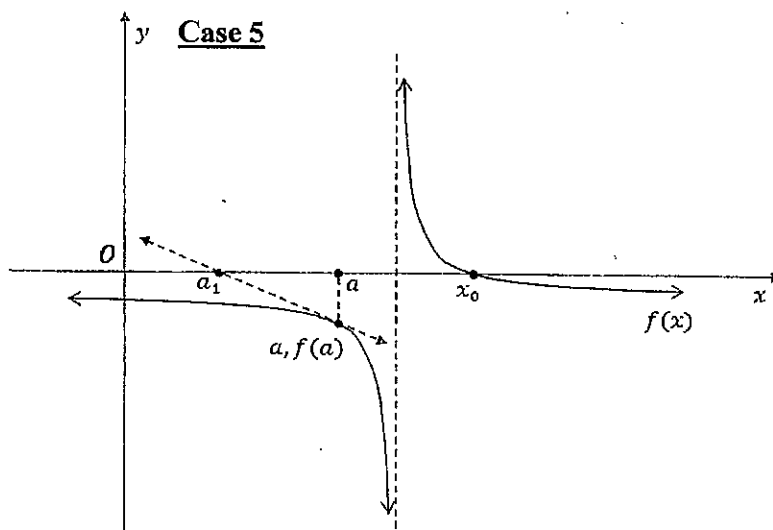
In cases 3, 4 and 5, Newton's method fails and  $a_1$  is not a better approximation than  $a$ .  
For cases 3 and 4, this is because  $f'(x) = 0$  at some point between  $a$  and  $x_0$  or at  $x = a$

(ii)

(1 mark)

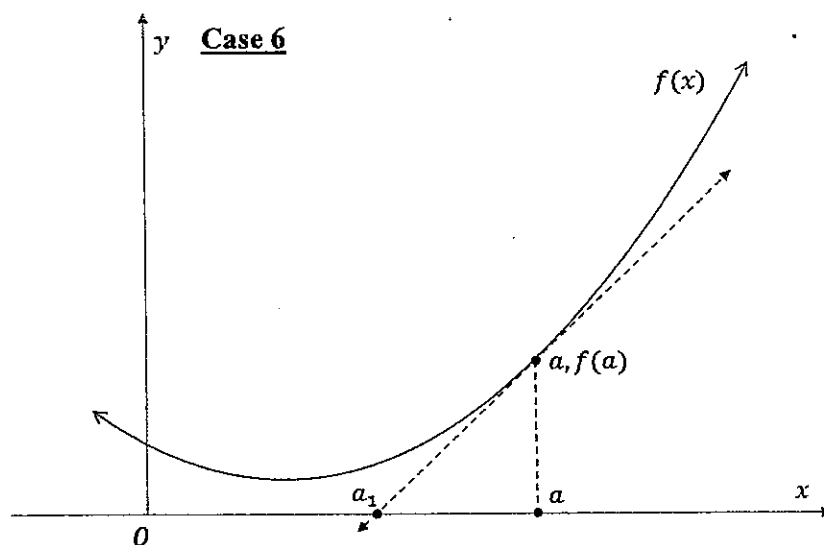


For case 5,  $f'(x)$  is undefined at some point between  $a$  and  $x_0$



**Question (4C)**

The original question asks to show a case where Newton's method does not apply "even though  $f(x)$  has a root near  $x = a$ ". Case 6 shows a function  $f(x)$  that does not have a root at all. Hence, this case is invalid as a solution.



Case 7 below shows a function similar to that in case 1. However, this case illustrates that the first approximation using Newton's method,  $a_1$  is not necessarily closer than the initial approximation  $a$ . However, subsequent approximations ( $a_2, a_3, \dots$ ) will get progressively closer to the required root  $x_0$ . This is not necessarily the case in cases 3, 4 and 5 where Newton's method fails.

